Palestine Technical University – Kadoorie

College of Engineering and Technology

Department of Computer Systems Engineering

Electronic voting

By:

Abrar Suliman - 202111014

Dalia Ghunim - 202110571

Dania Qasem - 202112505

Diana Hussain - 202110566

Supervisor:

Dr. Yazeed Saleet

Tulkarm, Palestine

June 2025

AKNOLGEMENT

In the name of Allah, the Most Gracious and the Most Merciful. We would like to take this opportunity to first and foremost thank God, the merciful for being our guide and for giving us the strength to write this thesis and carry on this project and for blessing us with many great people who have been our greatest support in both our professional and personal life.

We, the research team, dedicate this work to our dear parents, who supported and encouraged us in every step of our lives. To our friends who were with us in difficult times, and to our lecturers and instructors who inspired us with knowledge and belief in our abilities. We thank everyone who contributed to our success, and we hope that this research will be a valuable addition to the academic community.

We would like to extend our sincere thanks to our project supervisor, Dr. Yazeed Sleet, for his guidance, efforts and patience.

Special thanks to Dr. Yousef Daraghmeh, lecturer at Palestine Technical University - Kadoorie - for his time and effort spent to provide us with a better understanding of the requirements of such a project.

Finally, all thanks to all the group members for sharing the positivity and invaluable assistance.

ABSTRACT:

The integrity and transparency of electoral systems are critical for ensuring democracy and fairness in modern societies.

However, traditional voting systems, whether manual or electronic, often face challenges such as voter fraud, lack of privacy, and inefficiencies in vote counting.

This project proposes the development of a secure, blockchain-based electronic voting system that integrates facial recognition for voter authentication. The system aims to enhance voter confidence by employing advanced encryption algorithms and blockchain technology to ensure the confidentiality and immutability of votes while maintaining voter anonymity. By analyzing voting patterns and system performance, the project will also provide insights into the success rate and reliability of the implemented system.

This research seeks to address the limitations of existing voting systems, offering a more secure, transparent, and efficient alternative that can benefit policymakers, election authorities, and the public.

Ultimately, the study will contribute to the advancement of secure e-governance infrastructures.

Table OF Contents :

AKNOLGEMENT ii

ABSTRACT iii

TABLE OF CONTENT iv

LIST OF FIGURES vi

CHAPTER 1: INTRODUCTION 1

1.1 Background and Motivation 2

1.2 Problem Statement.....3

1.3 Objectives.....4

1.4 Overview.....5

CHAPTER 2: PRELIMINARY LITERATURE REVIEW 6

2.1

2.1.1 Introduction to Blockchain 7

2.1.2 Decentralized Voting Using Blockchain 8

2.1.3 Literature survey on Online Voting System Using Blockchain.....8

2.1.4 A Systematic Literature Review on Scalable Blockchain 9

2.1.5 A Survey of Blockchain Based on E-voting Systems.....10

2.1.6 A Survey on Smart Electronic Voting System Using Blockchain Technology....10

2.1.7 Survey on Voting System using Blockchain Technology.....11

2.2

2.2.1 Facial Recognition Technology.....12

2.2.2 Challenges.....13

2.2.3 Advantages Of Facial Recognition Technology.....14

CHAPTER 3: METHODOLOGY (PROCESS MODEL) 16

3.1 System Analysis and Requirements Gathering.....17

3.2 System Design.....18

3.3 Implementation.....18

3.4 Testing and Validation.....20

3.5 Deployment and Training.....20

3.6 Post-Election Analysis and Continuous Improvement.....21

3.7 Summary of Challenges and Solutions.....22

CHAPTER4: SOFTWARE DIAGRAMS.....23

4.1 SEQUANCE DIAGRAM 24

4.2 ACTIVITY DIAGRAM 25

4.3 USE CASE DIAGRAM.....29

4.4: ER DIAGRAM 30

CHAPTER5: SOFTWARE DIAGRAMS.....32

5.1 System Requirements 33

5.1.1 Functional Requirements ,,,,,,,,,,,,.....33

5.1.2 Non-Functional Requirements.....34

5.2 Role..... 35

CHAPTER6: SOFTWARE DIAGRAMS.....36

6.1 Overview of Implementation Stack 37

6.2 What is the Stack Used for Implementation?.....38

6.3 Integrated Development Environment 39

6.4 Front-End Technologies 39

6.5 Back-End Technologies.....40

6.6 Blockchain Architecture.....40

6.7 Facial Recognition Authentication.....41

6.8 Database and Hosting.....41

6.9 Organizational Tools.....42

REFERENCES

List of Figures

Figure 1 23

Figure 2 23

Figure 3 24

Figure 4 24

Figure 5 25

Figure 6 25

Figure 7 26

Figure 8 26

Figure 9 27

Figure 10 27

Figure 11 28

Figure 12 29

List Of Tables :

Table 1 21

Chapter 1

Introduction

1.1 Background and Context

Elections play a pivotal role in shaping governance and ensuring the representation of public opinion. However, globally, voting systems are plagued by challenges such as voter fraud, ballot tampering, and inefficiencies in manual vote counting. Even existing electronic voting systems often lack sufficient security mechanisms, leading to concerns about data breaches, vote manipulation, and voter privacy. These issues undermine public trust in electoral processes.

The advent of technologies such as blockchain and facial recognition presents an opportunity to revolutionize how elections are conducted. Blockchain technology, known for its immutability and decentralization, can ensure the integrity of the voting process by preventing tampering with votes. Simultaneously, facial recognition offers a robust mechanism to verify voter identities and prevent multiple votes, ensuring only eligible voters participate.

This project seeks to leverage these technologies to design and implement a secure, efficient, and transparent electronic voting system. By addressing existing vulnerabilities in traditional and electronic voting systems, this research aims to provide an innovative solution that ensures voter confidentiality, data security, and trustworthiness in elections.

1.2 Problem Statement

While modern technology has enabled the digitization of voting systems, many existing systems remain susceptible to fraud, privacy breaches, and inefficiencies. The primary issues this research addresses include:

- Voter Authentication: Difficulty verifying voter identity, which may lead to ineligible voting.
- Voting privacy and security: The risk of voter information being leaked or votes being manipulated, affecting the integrity of the election.
- Lack of transparency: There are no clear mechanisms to monitor the elections, which leads to a loss of confidence in the results.
- Human Error: Errors may occur during vote counting or data entry, affecting the accuracy of the results.

1.3 Aims and Objectives

- Develop a secure and scalable electronic voting platform that utilizes blockchain for vote storage and immutability.
- Integrate facial recognition for voter authentication to ensure election integrity.
- Implement advanced encryption algorithms to protect voter privacy and ensure the confidentiality of votes.
- Evaluate the system's performance through post-election analysis, including metrics such as voter turnout, system reliability, and user satisfaction.
- Provide recommendations for future adoption of the system in real-world electoral processes.

1.4 Overview

The proposed electronic voting system ensures secure, transparent, and efficient elections by integrating advanced technologies. Voters authenticate using a username, password, and facial recognition to verify their identity. A simple interface allows each voter to cast one vote, ensuring accessibility and preventing duplicate voting. The system uses encryption to protect data and blockchain to maintain anonymity and prevent tampering. Once the voting period ends, votes are securely tallied, and results are instantly available with full transparency. Detailed reports and analyses provide insights into voter turnout and system performance, modernizing the electoral process and building trust among stakeholders.

Chapter 2

LITERATURE REVIEW

2.1.1 Introduction to Blockchain:

is a distributed digital ledger technology that allows participants in a network to share and validate transactions in a secure and transparent manner without the need for intermediaries. The technology is designed to be decentralized, meaning that the data is stored on a network of computers instead of a central database. This makes it difficult to hack or manipulate the data, ensuring the integrity and security of the system. The blockchain technology gained popularity with the emergence of Bitcoin, which was the first decentralized cryptocurrency. However, the technology has since been applied to various industries, including finance, supply chain management, healthcare, and voting, among others. Blockchain works by creating blocks of data that are linked together in a chain, hence the name blockchain. Each block contains a unique code, known as a hash, that is generated based on the contents of the block. This hash is then used to link the block to the previous one, forming a chain of blocks. Once a block is added to the blockchain, it cannot be altered or deleted without the consensus of the network participants. This makes the technology immutable, ensuring that the data stored on the blockchain is tamper-proof and transparent. Overall, blockchain technology has the potential to revolutionize the way we store and share data, making it more secure, transparent, and accessible.

2.1.2 Decentralized Voting Using Blockchain:

A decentralized voting system built on the Ethereum blockchain has the potential to revolutionize the way we conduct elections. By leveraging the security, transparency, and immutability of blockchain technology, decentralized voting systems can eliminate many of the challenges and risks associated with traditional voting systems. In a decentralized voting system, each voter has a unique digital identity, and their vote is recorded on the blockchain, ensuring that the vote is tamper-proof and cannot be altered. Decentralized voting systems also eliminate the need for intermediaries, such as government agencies, to oversee the election process, making it more efficient and less susceptible to corruption or manipulation. Furthermore, decentralized voting systems can increase voter participation by allowing voters to cast their ballots from anywhere in the world, as long as they have an internet connection. This can lead to a more democratic and inclusive electoral process, with greater voter engagement and higher turnout. Overall, a decentralized voting system using the Ethereum blockchain has the potential to bring significant benefits to the electoral process, making it more secure, transparent, and accessible to everyone.

2.1.3 Literature survey on Online Voting System Using Blockchain:

Highly advanced security methods are necessary to introduce effective online voting system in the whole world. The aspect of security and transparency is a threat from global election with the conventional system. General elections still use a centralized system where one organization that manages it. Some of the problems that can occur in traditional electoral systems are with an organization that has full control over the database and system, it is possible to manipulate with the database. This paper presents a survey on some previous voting system that is used by different countries and organizations.

2.1.4 A Systematic Literature Review and Meta-Analysis on Scalable Blockchain Based Electronic Voting Systems:

Electronic voting systems must find solutions to various issues with authentication, data privacy and integrity, transparency, and verifiability. On the other hand, Blockchain technology offers an innovative solution to many of

these problems. The scalability of Blockchain has arisen as a fundamental barrier to realizing the promise of this technology, especially in electronic voting. This study seeks to highlight the solutions regarding scalable Blockchain-based electronic voting systems and the issues linked with them while also attempting to foresee future developments. A systematic literature review (SLR) was used to complete the task, leading to the selection of 76 articles in the English language from 1 January 2017 to 31 March 2022 from the famous databases. This SLR was conducted to identify well-known proposals, their implementations, verification methods, various cryptographic solutions in previous research to evaluate cost and time. It also identifies performance parameters, the primary advantages and obstacles presented by different systems, and the most common approaches for Blockchain scalability. In addition, it outlines several possible research avenues for developing a scalable electronic voting system based on Blockchain technology. This research helps future research before proposing or developing any solutions to keep in mind all the voting requirements, merits, and demerits of the proposed solutions and provides further guidelines for scalable voting solutions.

2.1.5A Survey of Blockchain Based on E-voting Systems:

Blockchain technology as a decentralized and distributed public ledger in a P2P network has recently gained much attention. In this technology, a linked block structure is applied, and a trusted consensus mechanism is established to synchronize data modifications, making it possible to develop a tamper-proof digital platform for data storage and sharing. We think that blockchain could be used in various interactive online systems, such as the Internet of Things, supply chain systems, voting systems, etc. The scope of this survey is to shed light on some recent contributions of the security and privacy issues associated with e-voting based on blockchain. At the end of this paper, we provided a comparison for the security and privacy requirements of the existing e-voting systems based on blockchain.

2.1.6A Survey on Smart Electronic Voting System Using Blockchain Technology:

India is the world's largest democracy with a population of more than 1 billion; India has an electorate of more than 668 million and covers 543 parliamentary constituencies. Voting is the bridge between the governed and government. The last few years have brought a renewed focus on to the technology used in the voting process. The current voting system has many security holes, and it is difficult to prove even simple security properties about them. A voting system that can be proven correct has many concerns. There are some reasons for a government to use electronic systems are to increase elections activities and to reduce the elections expenses. Still there is some scope of work in electronic voting system because there is no way of identification by the electronic voting system whether the user is authentic or not and securing electronic voting machine from miscreants. The proposed system is to develop a

compatible voting machine with high security by using Block-chain technology in order to increase security and transparency between the users.

2.1.7 Survey on Voting System using Blockchain Technology:

The use of information technology has in some ways revolutionized in many sectors. Evoting is said to be a symbol of modern democracy. While research on the topic is still emerging, it has mostly focused on the technical and legal issues instead of taking advantage of this technology and implementing it for good cause. Usefulness of e-voting will perform best when compared with the existing framework. The word Vote means to choose a candidate from a given list of candidates who will lead the organization or the group .The main goal of voting is to practice voting in such a way that every person votes to elect their Decentralized Voting System Using Ethereum Blockchain 5 leader. Most countries in the world, India is no exception, had trouble voting. Voting is still carried out in countries in physical mode. This physical mode process is not safe as it can be manipulated by members of voting commitment. There are many issues such as voting stations being too far and improper voting tools. The proposed flagship internet-based online voting system supported by blockchain technology solves this very problem. Blockchain technology uses encryption and hashing techniques with which it makes voting secure. In this case, each vote is considered as a unique transaction. A private blockchain is created using a peer to peer network where we store voting transactions. This application is programmed in such a way so that the details of voting are abstract from the user. Users will be given enough time for voting with the system running. The main purpose of this paper is to come up with a new unique solution, which does not require any technical skills. Since voting is in online mode, increased voter turnout is likely. In this project, the concept of developing an electronic voting system using blockchain technology is implemented.

2.2.1 Facial Recognition Technology :

Facial recognition technology is a biometric method used to verify an individual's identity by analyzing unique facial features. In a typical system, such as the one used for login, the process involves capturing a facial image and comparing it to a stored template for authentication. Here's a detailed technical breakdown of how facial recognition works in a login system:

1. Face Detection The first step involves detecting the face within an image or video stream. Computer vision algorithms, often based on deep learning (e.g., Convolutional Neural Networks, CNNs), are used to identify the location of a face. The system typically detects key facial landmarks such as the eyes, nose, mouth, and jawline.

2. Feature Extraction Once the face is detected, the next step is to extract distinctive features. These features include the distance between the eyes, the shape of the nose, the contour of the lips, and the overall geometry of

the face. These features are encoded into a numerical template, often referred to as a "face print" or "biometric template."

3. Template Creation The extracted features are then converted into a digital template, which represents the unique characteristics of an individual's face. This template is stored in a secure database. Importantly, the template is not an actual image but a series of measurements that mathematically encode the unique features of the face.

4. Face Matching (Verification) When a user attempts to log in using facial recognition, the system captures a new facial image and extracts the features from it in real-time. The newly created template is then compared to the stored template to check for a match. Modern facial recognition systems use various algorithms, such as cosine similarity or Euclidean distance, to quantify the similarity between the two templates.

5. Authentication Decision If the match is above a certain threshold, the system authenticates the user and grants access. In some systems, multiple templates may be used (e.g., multiple angles or images from different times) to improve accuracy. The comparison and decision making process happens very quickly, often in less than a second.

6. Security Measures Many modern facial recognition systems implement "liveness detection" techniques to prevent spoofing. These techniques ensure that the face presented is from a live person, not a photo or video. Liveness detection can involve analyzing subtle facial movements (e.g., blinking, head movement) or using infrared sensors to capture depth information of the face.

2.2.2 Challenges

- Environmental Factors: Lighting conditions, angle of the face, and facial expressions can affect the accuracy of recognition.

- Privacy Concerns: Since facial recognition involves the collection of biometric data, privacy and data security are critical issues.

- Bias and Accuracy: Some systems may exhibit performance biases depending on demographic factors like age, gender, and ethnicity, which need to be addressed with proper training and algorithm tuning.

2.2.3 Advantages Of Facial Recognition Technology :

1. High Accuracy and Reliability Facial recognition systems have become highly accurate due to advancements in deep learning algorithms and computer vision techniques. The technology analyzes key facial features (e.g., the distance between eyes, nose shape, and jawline) to create a unique biometric template for each individual. This provides a reliable method of authentication with lower false acceptance or rejection rates compared to traditional methods like passwords or PINs.

2. Non-Invasive and Contactless Authentication Unlike fingerprint or iris scanning, facial recognition is a contactless biometric method. It only requires the user to be in the line of sight of a camera, which can be advantageous in scenarios requiring hygiene or when users are not willing to touch physical devices. This also reduces user friction and enhances user experience by allowing hands-free authentication

3. Speed and Convenience Facial recognition can verify an individual in seconds, which makes it an efficient solution for authentication in a variety of applications, such as security systems, mobile devices, and access

control. Users don't need to remember passwords or carry cards, and the system can work seamlessly in the background for quick access

4. Scalability Facial recognition can scale easily across a large number of users without the need for physical hardware upgrades. Since facial templates can be stored digitally, it is easy to integrate the system with largescale databases, making it suitable for applications such as airport security, border control, and large organizations with high foot traffic.

5. Multi-factor Authentication Integration When paired with other authentication methods (e.g., passwords, smart cards, or behavioral biometrics), facial recognition enhances multi-factor authentication (MFA) systems, improving overall security without compromising convenience. Combining facial recognition with behavioral data such as typing patterns or walking gait can create even stronger user authentication systems.

6. Fraud Prevention Advanced facial recognition systems can include liveness detection features to ensure that the person presenting the face is physically present and not a static image or video. This prevents spoofing attempts and adds a layer of security against fraudulent behavior.

Chapter 3

METHODOLOGY

3.1 System Analysis and Requirements Gathering

Objective:

To understand the functional and non-functional requirements of the e-voting system and identify potential challenges. This phase involves consultations with stakeholders, including election authorities, cybersecurity experts, and potential voters.

Key Activities:

- Requirement Identification: Define the scope of the system, including user needs and regulatory compliance.
- Risk Analysis: Assess risks related to voter authentication, vote security, and system transparency.
- Technology Evaluation: Explore technologies such as biometric verification, blockchain, and encryption for system development.

3.2 System Design

Objective:

To create a robust architecture that addresses identified challenges while ensuring scalability, reliability, and user-friendliness.

Key Activities:

- Architectural Design:
  - Implement a modular system with layers for authentication, voting, and result analysis.
  - Use blockchain technology for vote recording to ensure transparency and immutability.
- Database Design:
  - Design a secure database to store voter information and voting data. Implement data encryption to protect sensitive information.
- User Interface Design:
  - Develop a user-friendly interface that accommodates voters of all technical skill levels.

3.3 Implementation

Objective:

To develop the e-voting system based on the design specifications using secure and scalable technologies.

Key Technologies:

- Biometric Authentication: To ensure only eligible voters can access the system and prevent duplicate voting.
- Encryption Algorithms: To secure communication between users and the server (e.g., AES for data encryption and RSA for key exchange).
- Blockchain: To create a tamper-proof voting ledger that guarantees vote integrity and transparency.

Steps to Address Specific Challenges:

1. Voter Authentication:
   - Use multi-factor authentication (e.g., biometric verification and unique voter IDs).
   - Prevent duplicate voting by cross-referencing voter credentials in real-time with a centralized database.
2. Vote Privacy and Security:
   - Encrypt all votes before transmission and store them in a blockchain ledger to prevent tampering.

- Use anonymization techniques to unlink votes from voter identities.

3. Transparency and Trust:

- Provide a public blockchain ledger for real-time monitoring of votes without compromising voter privacy.
- Enable independent audits of the voting process to ensure accuracy.

4. Post-Election Analysis:

- Integrate data analytics tools to generate reports on voter turnout, system reliability, and voting patterns.

3.4 Testing and Validation

Objective:

To ensure the system operates as intended under various scenarios and meets all security and usability requirements. Key Activities:

- Unit Testing: Test individual modules (e.g., voter authentication, encryption, blockchain integration).
- System Testing: Test the system as a whole for functionality, performance, and security.
- Simulated Elections: Conduct mock elections with a diverse group of users to validate usability and reliability.
- Penetration Testing: Identify and address vulnerabilities through ethical hacking.

3.5 Deployment and Training

Objective:

To deploy the system for real-world use and train stakeholders on its operation.

Key Activities:

- System Deployment: ○ Deploy the system on secure servers with redundant backups.
  - ○ Ensure 24/7 monitoring during election periods.
- User Training:
  - Train election authorities and technical support teams on system usage and troubleshooting.
  - Provide voters with guides and tutorials on how to use the system.
- Public Awareness Campaign:
  - Launch a campaign to educate voters about the benefits and security features of the e-voting system.

3.6 Post-Election Analysis and Continuous Improvement

Objective:

To analyze the performance of the system, gather feedback, and implement improvements. Key Activities:

- Data Analysis:
  - Use built-in analytics tools to assess voter turnout, system reliability, and processing times.
- Feedback Collection:
  - Conduct surveys with voters and election authorities to identify potential areas for improvement.
- System Updates:
  - Address any issues identified during post-election analysis and

3.7 Summary of Challenges and Solutions

Table 1 : summary

Table 1 : summary

CHALLENGE	PROPOSED SOLUTION
VOTER AUTHENTICATION	Implement biometric and multi-factor authentication to ensure only eligible voters can cast their votes.
VOTE PRIVACY AND SECURITY	Use encryption and blockchain to protect votes from tampering and ensure voter anonymity.
TRANSPARENCY AND TRUST	Provide a public blockchain ledger and enable independent audits to build trust in the system.
POST-ELECTION ANALYSIS	Integrate analytics tools to evaluate voter turnout, system reliability, and overall election success.

Chapter 4

SOFTWARE DIAGRAMS

4.1 Sequence Diagram

4.1.1 Register Sequence Diagram

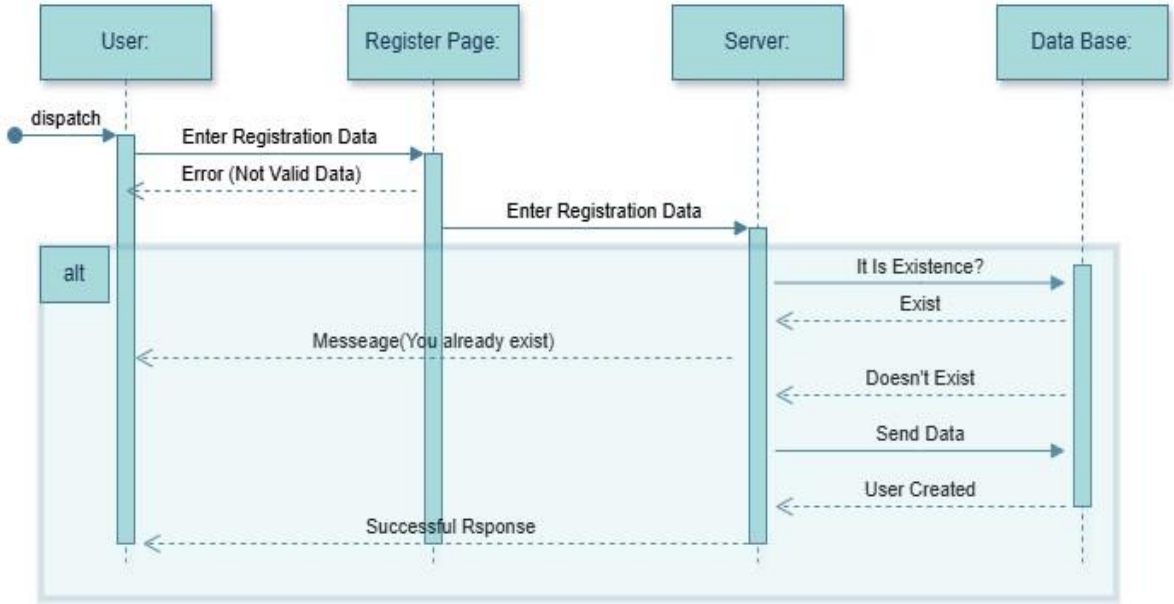

FIGURE 1 : Register Sequence Diagram

4.1.2 Login Sequence Diagram

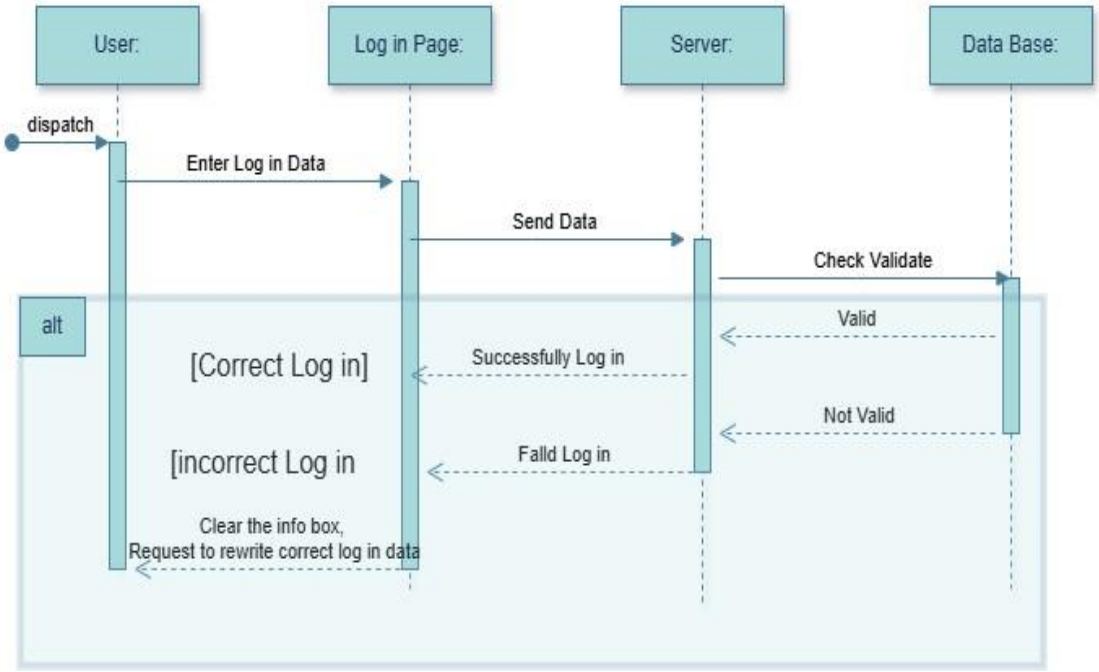

FIGURE 2 : Login Sequence Diagram

4.1.3 Vote Sequence Diagram

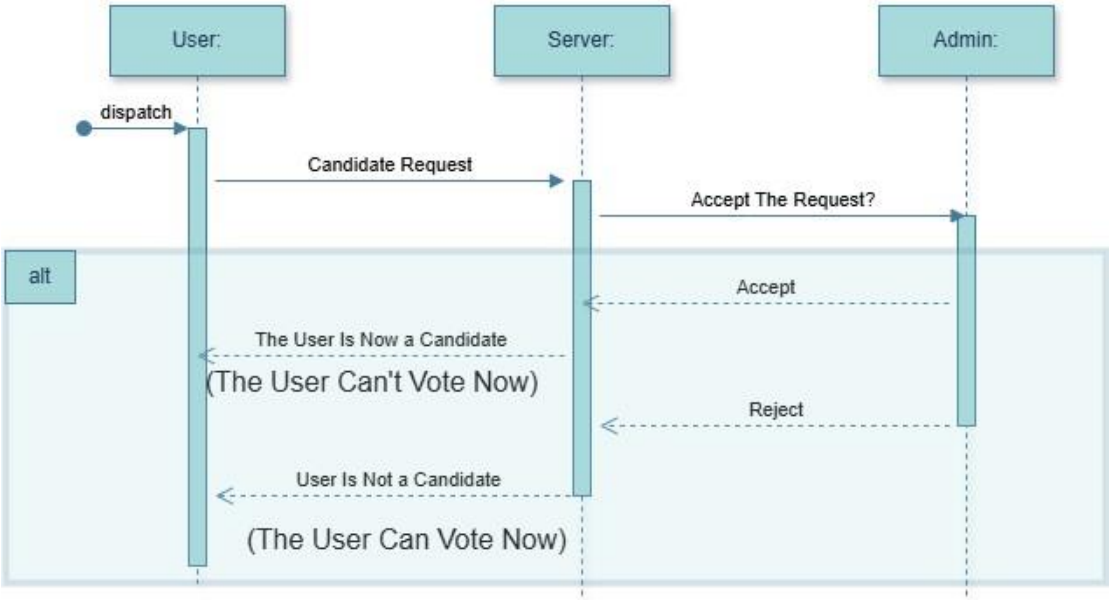

FIGURE 3 : Vote Sequence Diagram

4.1.4 Candidate Sequence Diagram

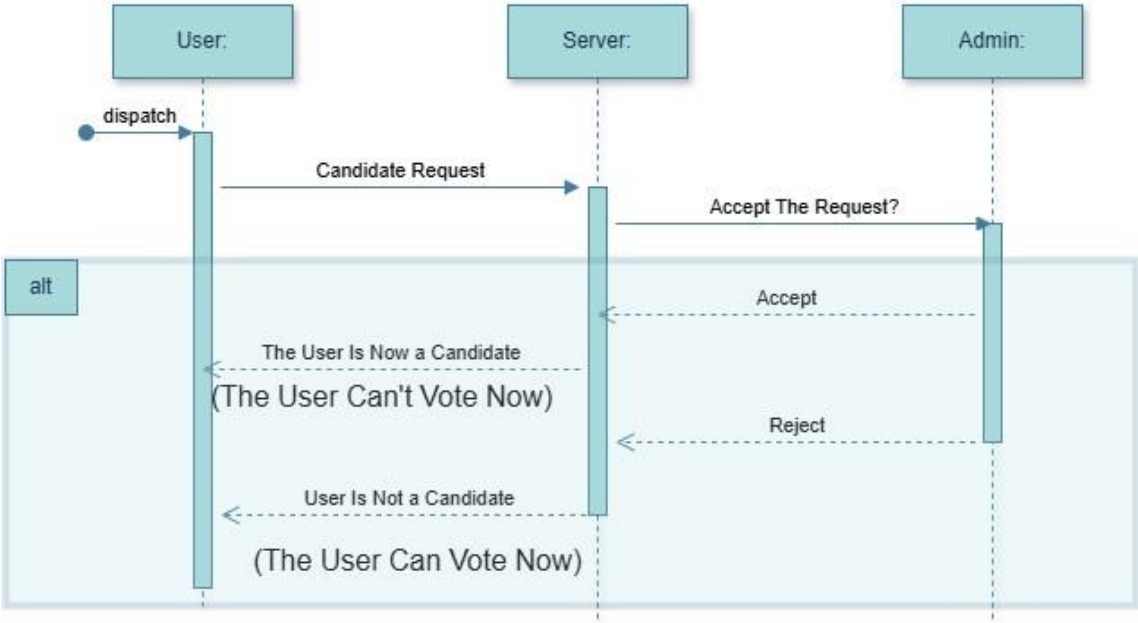

FIGURE 4 :Candidate Sequence Diagram

4.2 Active Diagram

4.2.1 Login Active Diagram

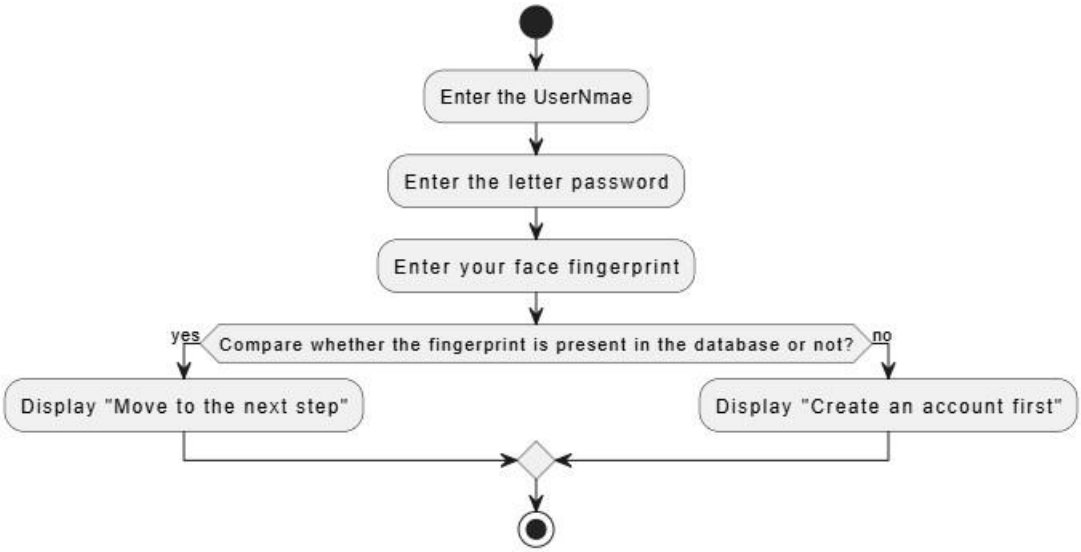

FIGURE 5 :Login Active Diagram

4.2.2 Register Active Diagram

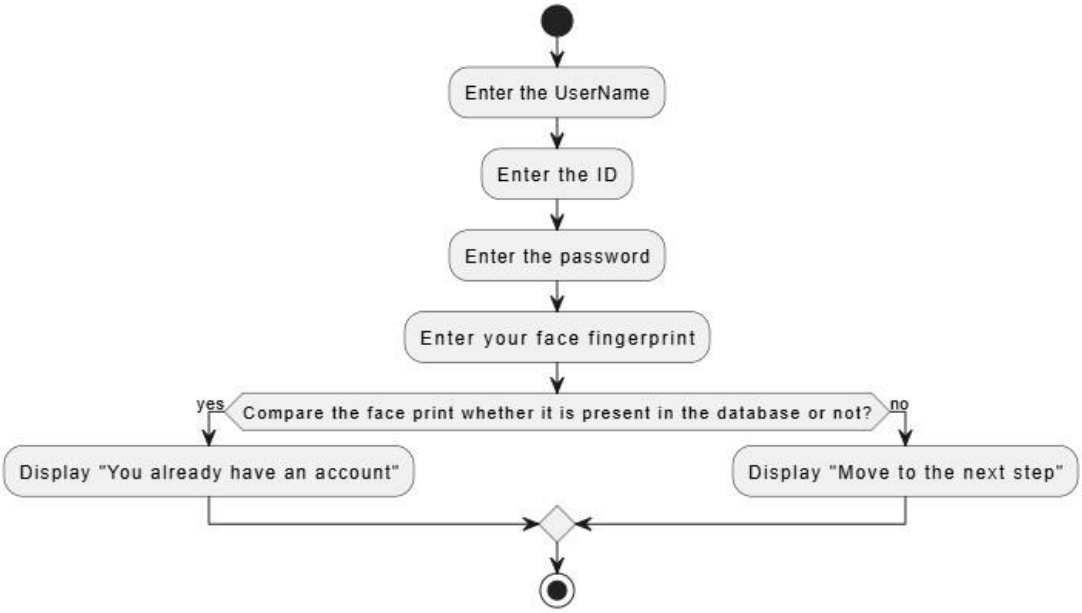

FIGURE 6 :Register Active Diagram

4.2.3 Handing User Active Diagram

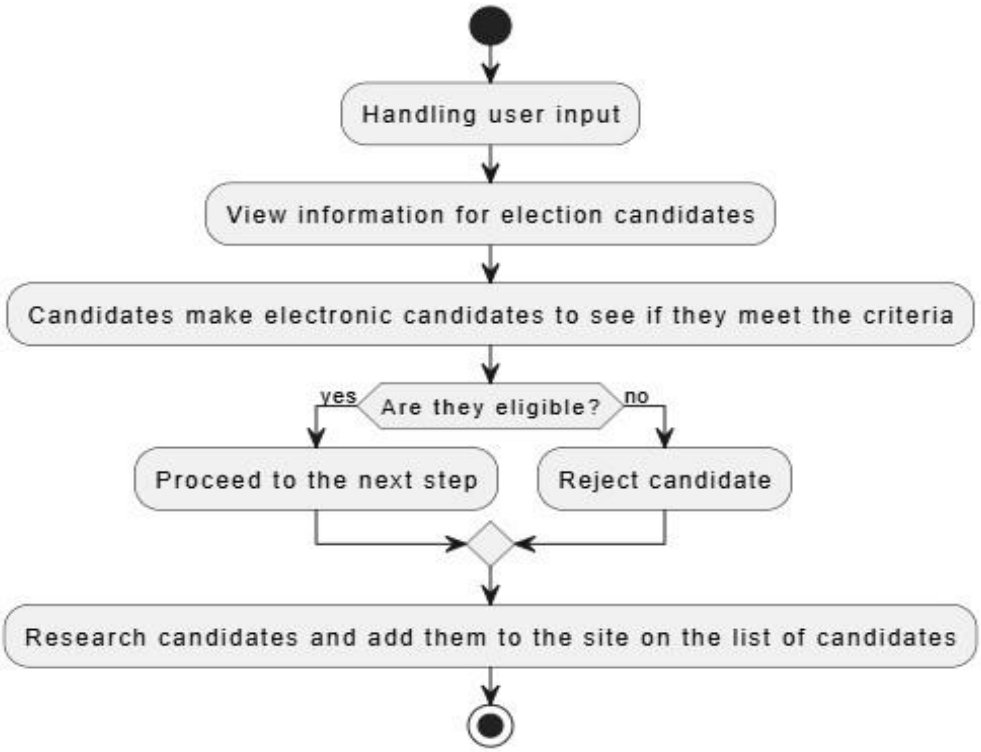

FIGURE 7 :Handing user Active Diagram

4.2.4 Nominating for elections Active Diagram

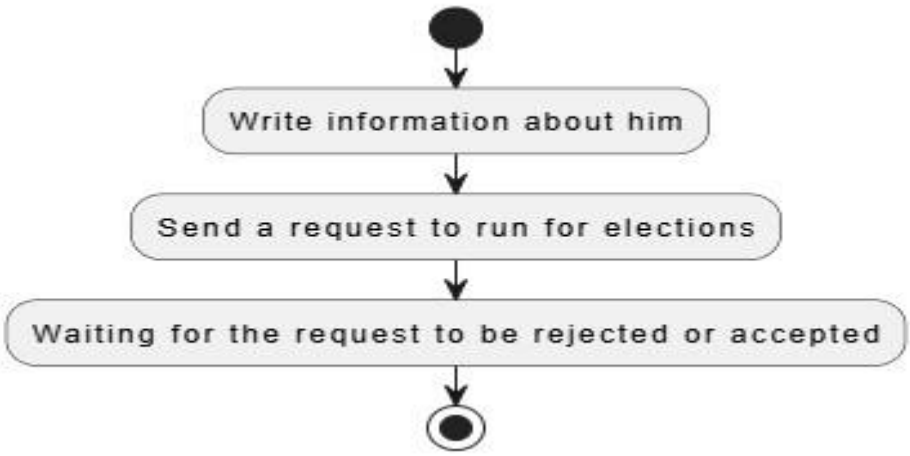

FIGURE 8 : Nominating for elections Active Diagram

4.2.5 Create New Admin Active Diagram

FIGURE 9 : Create new admin Active Diagram

2.4.6 Vote Active Diagram

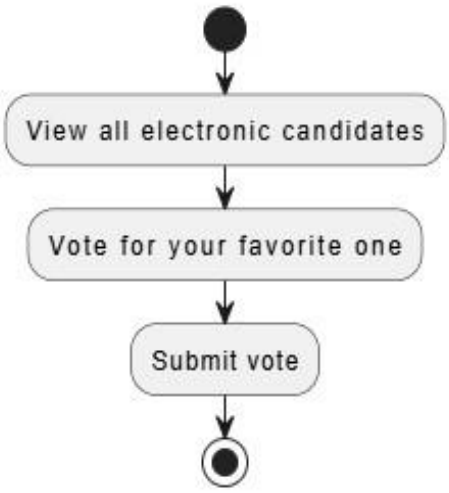

FIGURE 10 : Vote Active Diagram

4.3 Use Case Diagram

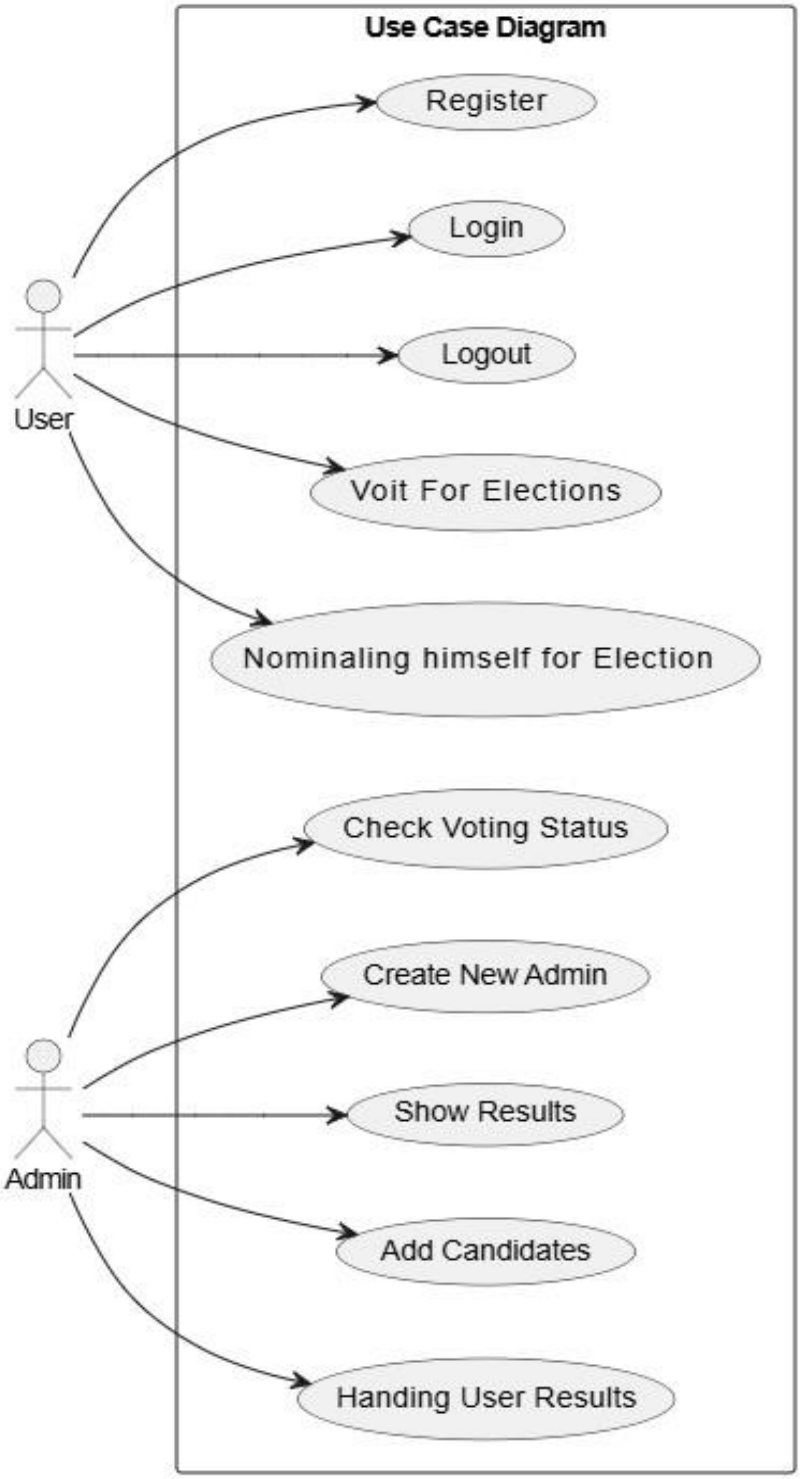

FIGURE 11 : Use Case Diagram

4.4 ER-Diagram

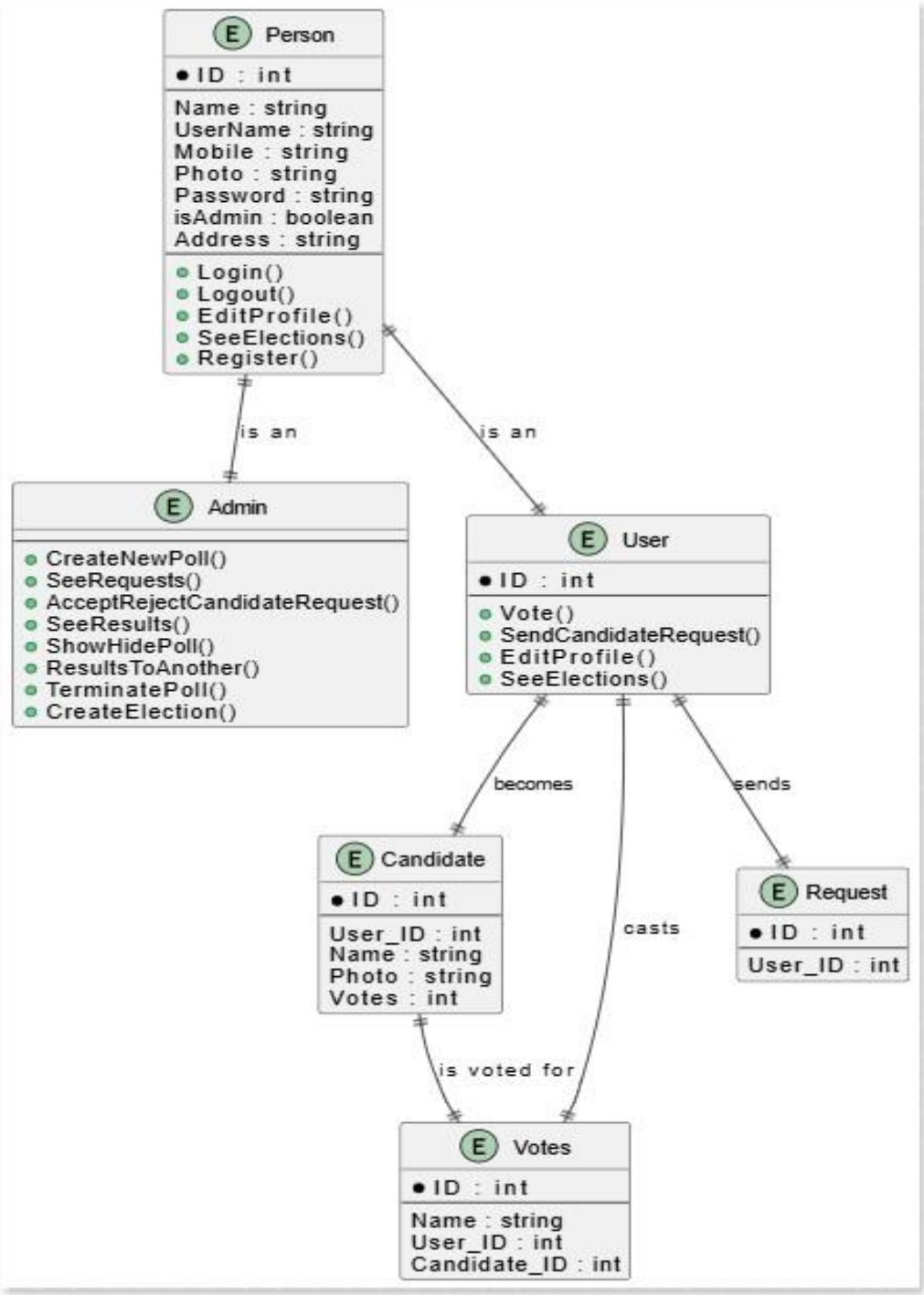

FIGURE 12 : ER- Diagram

Chapter 5 :

System Requirements

5.1 Functional Requirements

User Registration & Login

The system shall allow users to log in using a username and password.

The system shall verify user identity using facial recognition (Face ID).

The user's facial data shall be securely encrypted and stored on the blockchain.

Admin Panel

The system shall allow the admin to create a new election.

When creating an election, the admin shall be able to specify the end date/time of the election.

The admin shall be able to add candidates to the election.

The admin shall be able to end or cancel the election at any time before the scheduled end.

The admin shall be able to view the status of the election and the number of voters.

Voting Process

Users shall be allowed to vote only during the active election period, based on the configured start and end time.

Each user shall be allowed to cast only one vote per election.

Face ID Verification

The system shall use facial recognition to match the user with their stored biometric data.

Login attempts that fail facial recognition shall be rejected.

Election Results

The system shall automatically calculate and display the election results after the election ends.

The candidate with the highest number of votes shall be declared the winner.

Results shall be final and immutable after the election is completed.

5.1.2Non-Functional Requirements

Security

All sensitive information, including passwords and facial data, shall be encrypted.

The system shall use secure communication protocols (e.g., HTTPS).

The system shall protect against common security threats such as SQL injection and replay attacks.

Scalability

The system shall support a large number of concurrent users, especially during peak voting times.

Availability & Reliability

The system shall have an availability rate of at least 99.9%.

Regular backups shall be implemented to prevent data loss.

Performance

Login and voting operations shall respond within 2 seconds under normal conditions.

Face ID verification shall complete within 5 seconds

Usability

The system shall have a user-friendly interface ...

5.2 Role

1. Admin (Administrator)

The admin is responsible for managing the overall election system. This role includes full control over election creation and monitoring. Responsibilities include:

Creating new elections.

Setting the start and end date/time of each election.

Adding and managing candidates.

Ending or canceling ongoing elections.

Viewing and managing voter activity.

2. Voter (User)

The voter is a regular user of the system whose main role is to participate in elections. Responsibilities include:

Logging into the system using username, password, and Face ID.

Verifying identity via facial recognition.

Viewing available and active elections.

Casting a vote for one candidate per election.

Viewing the election results after completion.

3. Candidate

A candidate is a user nominated to run in a specific election. Responsibilities include:

Being visible to voters as a choice in the election.

Optionally accessing a limited view of their status (e.g., current vote count if permitted).

Chapter 6 :

Technologies and Implementation

6.1 Overview of Implementation Stack

- Integrated Development Environment

  Visual Studio Code

- Front-End Technologies
  - React.js
  - Ant Design (AntD)
  - Axios

- Back-End Technologies
  - Node.js
  - Express.js

- Blockchain Technologies
  - Ganache
  - Truffle

- Facial Recognition
  - FaceApi.js
- Database
  - MongoDB

- Deployment & Hosting
  - Amazon Web Services (AWS)

- Organizational & Collaboration Tools
  - GitHub – Version control
  - WhatsApp – Team communication

6.2 What is the Stack Used for Implementation?

The eVote system leverages a modern web development stack that focuses on security, scalability, and ease of use. The selected tools and frameworks were chosen for their open-source nature, robust community support, and ability to handle real-time, high-integrity transactions required in a digital voting system.

- **React.js** was selected as the front-end framework due to its modularity, performance, and developer efficiency. Combined with Ant Design, it enabled the creation of a clean, responsive user interface.

- **Node.js** and **Express.js** formed the server architecture, offering a scalable and fast environment for handling user requests and backend logic.

- **Truffle** and **Ganache** provided a local Ethereum development environment for deploying and testing blockchain smart contracts securely.

- **FaceApi.js**, a JavaScript-based face recognition library, enabled secure and accurate facial authentication within the browser.

- **MongoDB**, a NoSQL database, was used to store user data and voting logs in a flexible and scalable format.

- **AWS** hosted the application, offering reliability and deployment flexibility

6.3 Integrated Development Environment

6.3.1 Visual Studio Code

Visual Studio Code (VS Code) was used throughout development due to its lightweight structure, rich plugin ecosystem, and strong support for JavaScript, Node.js, and React. It enabled seamless collaboration with GitHub integration, efficient debugging, and code linting features.

Why VS Code?

- Cross-platform and open-source.
- Built-in terminal and Git integration.
- Extensions for React, Node, and Solidity (for smart contracts).
- Real-time error highlighting and IntelliSense

6.4 Front-End Technologies

6.4.1 React.js

React.js, developed by Meta, is a component-based JavaScript library for building interactive UIs. It allowed us to break the interface into reusable pieces, improving maintainability and reducing development time.

6.4.2 Ant Design (AntD)

Ant Design is a popular React UI framework offering pre-built components that follow modern UX/UI standards. It ensured consistency and helped rapidly build professional-grade interfaces.

6.4.3 Axios

Axios is a promise-based HTTP client for JavaScript. It was used to handle communication between the frontend and the backend, simplifying the handling of asynchronous API requests and responses.

6.5 Back-End Technologies

6.5.1 Node.js & Express.js

Node.js is a fast, event-driven runtime built on Chrome's V8 engine. With Express.js, a lightweight web framework, we structured the backend logic, API routing, and middleware in an efficient and scalable way.

Key Benefits:

- Asynchronous I/O for scalability.
- Robust middleware and routing.
- Fast development cycles with NPM modules.

6.6 Blockchain Architecture

6.6.1 Ganache

Ganache is a local Ethereum blockchain used to simulate the deployment of smart contracts. It allowed us to test blockchain functionality in a safe environment without real costs.

6.6.2 Truffle

Truffle is a development framework for Ethereum. It provided tools for writing, compiling, and deploying smart contracts, as well as interacting with them through JavaScript, which integrated seamlessly with our Node.js backend.

Use Cases:

- Voting logic and immutability via smart contracts.
- Transparent result verification using on-chain records.

6.7 Facial Recognition Authentication

6.7.1 FaceApi.js

FaceApi.js is a powerful face detection and recognition library built with TensorFlow.js. It runs directly in the browser, eliminating the need to send sensitive biometric data to external servers.

Why FaceApi.js?

- Lightweight and privacy-focused (runs client-side).
- Compatible with most modern browsers.
- High accuracy in face detection and matching.

6.8 Database and Hosting

6.8.1 MongoDB

MongoDB is a NoSQL, document-oriented database. It was chosen for its flexible schema design and ability to handle high volumes of structured and unstructured data.

Why MongoDB?

- Easy integration with Node.js.
- High performance for read/write operations.
- JSON-like structure (BSON) ideal for JavaScript-based apps.

6.8.2 Amazon Web Services (AWS)

The system was deployed on AWS for its scalability, global infrastructure, and secure hosting environment. AWS allowed us to deploy both backend services and the front-end app efficiently

6.9 Organizational Tools

6.9.1 GitHub

GitHub served as our version control and collaboration platform. It helped track changes, manage branches, and maintain code integrity throughout the development process.

6.9.2 WhatsApp

WhatsApp was used for real-time team communication and coordination. It allowed for quick discussions, troubleshooting, and updates, especially under tight timelines.

REFERENCES

- Anasune, V., Choudhari, P., Kelapure, M., Shirke, P., & Halgaonkar, P. (2025). Decentralized Voting System Using Blockchain. GitHub. Retrieved from https://github.com/Krish-Depani/Decentralized-Voting-System/blob/main/Project%20Report%20github.pdf

- OpenAI. (2025). ChatGPT conversation. Retrieved January 2, 2025, from https://chat.openai.com
- https://www.mongodb.com/docs/?msockid=1b9ec634cf34630a3db2d329cea46233
- https://archive.trufflesuite.com/docs/
- https://github.com/justadudewhohacks/face-api.js/
- https://react.dev/learn
- https://expressjs.com/